

Math 550, S20
Quiz 6

Name: Answer

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) Find the local maximum and minimum values and saddle points of the function $f(x, y) = x^4 + y^4 - 4xy + 2$.

Solution: $Df(x, y) = [f_x \ f_y] = [4x^3 - 4y \ 4y^3 - 4] = 0$

$$\begin{cases} 4x^3 - 4y = 0 \\ 4y^3 - 4x = 0 \end{cases} \Rightarrow \begin{cases} x^3 - y = 0 \\ y^3 - x = 0 \end{cases} \Rightarrow \begin{cases} x^3 - x = 0 \\ y^3 - y = 0 \end{cases} \Rightarrow \begin{cases} x(x^2 - 1) = 0 \\ y(y^2 - 1) = 0 \end{cases}$$

$$\Rightarrow \begin{matrix} x=0 & x=1 & x=-1 \\ \downarrow & \downarrow & \downarrow \\ y=0 & y=1 & y=-1 \end{matrix} \Rightarrow \text{critical points } (0,0), (1,1), (-1,-1)$$

Next, $Hf(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{bmatrix}$

For $(0,0)$, $Hf(0,0) = \begin{bmatrix} 0 & -4 \\ -4 & 0 \end{bmatrix}$, $\Rightarrow d_1 = 0, d_2 = -16 < 0$
 $\Rightarrow (0,0)$ is a saddle point.

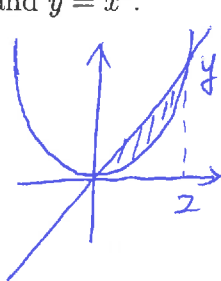
For $(1,1)$, $Hf(1,1) = \begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix}$, $\Rightarrow d_1 = 12 > 0, d_2 = 12 > 0$
 $\Rightarrow (1,1)$ is a local minimum with $f(1,1) = -1$.

For $(-1,-1)$, $Hf(-1,-1) = \begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix}$, $\Rightarrow d_1 = 12 > 0, d_2 = 12 > 0$
 $\Rightarrow (-1,-1)$ is a local minimum with $f(-1,-1) = -1$.

Problem 2. (3 points) Calculate the iterated integral $\int_0^1 \int_1^2 (4x^3 - 9x^2y^2) dy dx$

$$\begin{aligned} \int_0^1 \int_1^2 4x^3 - 9x^2y^2 dy dx &= \int_0^1 [4x^3y - 3x^2y^3]_{y=1}^{y=2} dx \\ &= \int_0^1 [(8x^3 - 24x^2) - (4x^3 - 3x^2)] dx \\ &= \int_0^1 4x^3 - 21x^2 dx \\ &= [x^4 - 7x^3]_{x=0}^{x=1} = (1 - 7) - 0 = -6 \end{aligned}$$

Problem 3. (3 points) Evaluate $\iint_D x + 2y \, dA$, where D is the region enclosed by $y = 2x$ and $y = x^2$.



$$\begin{aligned} \begin{cases} y=2x \\ y=x^2 \end{cases} &\Rightarrow x^2=2x \\ &\Rightarrow x^2-2x=0 \Rightarrow x(x-2)=0 \\ &\quad x=0 \text{ or } x=2 \\ &\quad \downarrow \quad \downarrow \\ &\quad y=0 \quad y=4 \end{aligned}$$

$$D = \{(x, y) \mid 0 \leq x \leq 2, x^2 \leq y \leq 2x\}$$

$$\begin{aligned} \iint_D x + 2y \, dA &= \int_0^2 \int_{x^2}^{2x} x + 2y \, dy \, dx \\ &= \int_0^2 \left[xy + y^2 \right]_{y=x^2}^{y=2x} dx \\ &= \int_0^2 [(2x^2 + 4x^2) - (x^3 + x^4)] dx \\ &= \int_0^2 -x^4 - x^3 + 6x^2 dx \\ &= \left(-\frac{x^5}{5} - \frac{x^4}{4} + 2x^3 \right) \Big|_{x=0}^{x=2} \\ &= -\frac{32}{5} - \frac{16}{4} + 16 = \frac{28}{5} \end{aligned}$$